Andrew Koja

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Education

Michigan State University Aug 2024 – May 2028
Bachelor of Science in Computer Science *East Lansing, MI*

- **GPA:** 3.53 Cumulative, 3.83 Major
- **Relevant Coursework:** Algorithms & Data Structures *(In Progress)*, Computer Organization & Architecture *(In Progress)*, Programming II (C++), Discrete Structures, Programming I (Python), Calculus I–III

Experience

Apostles4Jesus June 2023 – Present
Full-Stack Software Developer *Macomb, MI*

- Designed and deployed a bilingual (English/Arabic) podcast platform for an organization with 15,000+ social media followers using HTML/CSS, JS, and Netlify
- Mitigated OWASP Top 10 risks with CSP, rate limiting, input sanitization, and reCAPTCHA via serverless Netlify Functions
- Implemented WebP conversion and lazy loading to cut homepage payload 10x (3.3MB to 336KB), improving load times for mobile users on slow connections

Projects

Raquette | *Python*, *PyTorch*, *OpenCV*, *FastAPI*, *React*, *Docker* raquette.vercel.app

- Architected a 6-stage tennis video analysis pipeline behind an async FastAPI job service with progress polling and range-request video streaming to a React frontend
- Halved processing time on a 2-minute rally (1,228s to 589s), cutting court detection from 640s to 96s by rejecting wrong-camera court fits and backing off the line-fitting search on failures; outputs verified unchanged
- Deployed a Dockerized backend (Hugging Face Spaces) and frontend (Vercel) with demo limits and 24-hour data retention; wrote a reproducible eval script (94% ball-detection recall)

LightAPI | *React*, *FastAPI*, *PyTorch*, *SQLite*, *WebSockets*, *Docker* lightapi.dev

- Engineered a searchable directory of 1,500+ public APIs where any entry becomes a live monitor in one click, recording 28,800+ latency readings daily across 10 endpoints
- Trained per-endpoint PyTorch LSTMs on each service's latency history to flag anomalies (z-score fallback under 100 readings), and built root-cause correlation linking regressions to GitHub deploy webhooks
- Shipped a Python SDK with a non-blocking `@monitor` decorator for latency tracing, backed by a containerized FastAPI ingest service

FootyLive | *Python*, *FastAPI*, *React*, *TypeScript*, *SQLite*, *WebSockets*, *Redis* FootyLive

- Built a real-time soccer dashboard for 12 competitions, delivering live scores every 30 seconds over WebSockets with polling fallback
- Developed broadcast-style knockout brackets for the Champions League, Euros, and World Cup with two-leg aggregation and penalty-shootout resolution
- Maintained uptime via background sync, in-memory caching, and season archiving that serves finished seasons without external API calls

CrimeScope | *React*, *Vite*, *Leaflet*, *Zustand*, *Recharts*, *SODA / ArcGIS APIs* crime-scope.vercel.app

- Created a crime dashboard (pin, cluster, heatmap) filtering 50,000+ records client-side across 17 city APIs
- Added shareable URL state and stale-data detection across 17 open-data portals

Technical Skills

Languages & Frameworks: Python, JavaScript, TypeScript, C++, HTML/CSS, SQL, FastAPI, React, Tailwind CSS, Node.js
Tools & DevOps: Git/GitHub, GitHub Actions (CI/CD), Docker, pytest, SQLite, Redis, Vercel, Claude Code, Cursor, Codex
ML & AI: PyTorch, scikit-learn, CatBoost, OpenCV, Computer Vision, Anomaly Detection
Concepts: Real-Time Systems, REST APIs, Data Visualization, Server-Side Caching